

A comment on “Effects of Voice Pitch on Social Perceptions Vary With Relational Mobility and Homicide Rate”

Abstract

The study by Aung et al. (2024) analysed the influence of voice's fundamental frequency (f_0) on perceptions of individual traits such as attractiveness, prestige and formidability. Authors highlight the influence of the socioecological factors of relational mobility and violence. Using data from 22 countries and Bayesian mixed effect models, authors report that low male f_0 are associated with higher perceptions of formidability and prestige in contexts of high mobility and homicide rates. The study shows that high female f_0 is related to women's perceptions of flirtatiousness in contexts of lower relational mobility where competition for mates is larger. We successfully reproduced the main findings of the study and performed an analysis using an alternative frequentist approach.

1. Introduction

Data were obtained from 3173 participants in 44 locations across 22 nations. Participants listened to pairs of voices and were asked to choose one voice from each pair (first or second) in response to specific questions. Men were asked to indicate which male voice in each pair sounded more respected, admired, talented, and successful (“prestige”) and which male voice in each pair sounded more likely to win a physical fight (“formidability”). Men were also asked to indicate which female voice in each pair sounded more attractive for a short-term, uncommitted romantic relationship (“short term”) and which female voice in each pair was more attractive for a long-term, committed relationship such as steady dating or marriage (“long term”). Women were asked to indicate which female voice in each pair would be more attractive to men (“attractive”) and which female voice in each pair sounded more interested in attracting men (“flirtatious”). Women were also asked to indicate which male voice in each pair sounded more attractive for a short-term, uncommitted romantic relationship (“short term”) and which male voice in each pair was more attractive for a long-term, committed relationship such as steady dating or marriage (“long term”). For the data analysis, binary logistic Bayesian mixed-effects models were fit to the data using RStan via the brms package.

In the present report, we test whether the main results and claims made by the authors are computationally reproducible. We focus on the analysis of the impact of the fundamental frequency (f_0) on several indicators of perceptions of female and male voices, as reported in the text (pp. 255-257).

2. Computational Reproducibility:

We used the replication package provided by the authors here: <https://osf.io/tnygr/>. The replication package lacked a README file to guide readers. The authors provide raw data and distance matrices, as well as analysis data that results from running the Bayesian models. Very minimal cleaning code is used, but this can be explained by the experimental nature of this study and cannot be deemed a limitation.

Authors specify in the text particular versions of R packages used for analysis. We tried using the same versions; some of them were hard to access and install. Running the models was ultimately successful, but a lot of set-up and troubleshooting was necessary on some of our machines (we used both Windows and macOS devices).

We noticed a minor error when checking the code. At the start of the main code file ("PsychSci.R"), data is imported as a `data.frame` object called "data"; however, the subsequent model uses an object called "newdata", which is never defined anywhere else. We believe that "newdata" was actually supposed to be "data", and making this change solved the issue.

The authors provided code for over 70 models that they ran. We chose to focus on models M1, M17, M18, M19, M20, M23, M57, M58, and M62, since they seemed to constitute the most significant part of the authors' claims. We selected these models as opposed to reproducing all models due to computational and time constraints. We compared models that resulted from the code and analysis data provided to find whether they were equal. Despite some factors that made our reproduction more cumbersome, we ultimately consider that the authors fulfilled all the reproducibility criteria (see Table 1).

Table 1. Reproducibility criteria

	Fully	Partial	No
Raw data provided	x		
Cleaning code provided	x		
Analysis data provided	x		
Analysis code provided	x		
Reproducible from raw data	x		
Reproducible from analysis data	x		

Authors reported coefficients, estimated marginal means and post-hoc comparisons. Results reported in text are a mix of *beta* (*b*) coefficients and odds ratios (OR); however, the labels were often incorrect. This made the comparison of results in text and those obtained when running the models confusing and time-consuming. Part of the *b* coefficients reported in the text are actually odds ratios. This has been checked by exponentiating the coefficients from the model output (e^b) and comparing them to those reported in the article (see Table 2). While in those instances the coefficients match after exponentiating, many of the credible intervals (CrI) still don't match – we are not sure why that is. The differences are small and don't affect the claims made by the authors, but they go beyond what could be explained by rounding error. We also noticed some very minor inconsistencies that we suspect come from at times truncating and at times rounding up results to three or two decimals.

3. Discrepancies between pre-registered report and article:

While we appreciate the authors' commitment to pre-registration and open-science practices, we observed a possible shift in emphasis between the pre-registration form and the final manuscript. For instance, the pre-registered report included health, inequality, and food security as key factors, whereas the final manuscript focuses primarily on the

role of relational mobility and male rivalry without linking the two to the key factors highlighted in the pre-registered report.

4. Findings on the moderating role of relational mobility

The authors did not report findings related to the moderating roles of male rivalry or relational mobility. Our analysis indicates that relational mobility does not moderate women's preference for lower male voice pitch. Instead, women generally prefer a higher male pitch across varying degrees of male rivalry and relational mobility. A higher male voice pitch (f_0) slightly increased the probability of short-term or long-term partner choice judgments across nations, contrary to theoretical expectations.

All in all, the reported results show a slight increase in the probability of short-term partner choice judgments for higher male voice pitch, contrary to theoretical expectations. No strong evidence was found linking pitch to long-term mate preferences, suggesting a minimal role for pitch in shaping preferences across diverse cultures.

5. Robustness reproduction:

As a robustness check, we tried to reproduce the main findings of this paper using a frequentist modelling approach. We used logistic regression mixed models and included all the variables used by the authors, except for geographic relatedness. Results can be found in the replication table. Overall, the results from the frequentist models are very similar to the results from the Bayesian models and do not change the interpretation advanced by the authors (see Table 2).

6. Conclusions:

In this brief report, we presented our findings after attempting to replicate the results of the original authors and implementing an additional robustness check. We find that the main findings could be reproduced successfully and that they hold even when using a modelling approach based on the frequentist framework. While we did find some issues related to the code provided, the reporting of the coefficients, and some inconsistencies between the preregistration and the article, they were minor and did not clash with the overall narrative of the paper.

Table 2. Replication results

	Indicator	Coefficients reported in text	Coefficients from model output	Coefficients from model output in odds ratios	Frequentist model coefficients	Model source
Men's perception of male voices	Formidability	$b = 2.03$, 89% CrI: [1.55, 2.66]	$b = 0.711$, 89% CrI: [0.278, 1.031]	OR = 2.030, 89% CrI: [1.321, 2.804]	$b = 0.785$, 95% CI: [0.666, 0.905]	M1
	Prestige	$b = 1.93$, 89% CrI: [1.48, 2.56]	$b = 0.662$, 89% CrI: [0.223, 0.977]	OR = 1.939, 89% CrI: [1.249, 2.657]	$b = 0.735$, 95% CI: [0.615, 0.854]	M1
	Formidability vs prestige contrast	OR = 0.95, 89% CrI: [0.89, 1.01]	OR = 0.957, 89% CrI: [0.881, 1.035]	-	OR = 0.955, 95% CI: [0.841, 1.086]	M1
	Formidability vs short term attractiveness	OR = 0.77, 89% CrI: [0.71, 0.83]	OR = 0.771, 89% CrI: [0.702, 0.842]	-	OR = 0.769, 95% CI: [0.677, 0.874]	M1
	Formidability vs long term attractiveness	OR = 0.91, 89% CrI: [0.84, 0.98]	OR = 0.909, 89% CrI: [0.825, 0.991]	-	OR = 0.908, 95% CI: [0.799, 1.033]	M1
	Formidability * RM	$b = 1.57$, 89% CrI: [1.11, 2.29]	$b = 0.450$, 89% CrI: [0.098, 0.831]	OR = 1.568, 89% CrI: [1.103, 2.294]	$b = 0.394$, 95% CI: [0.072, 0.715]	M19
	Prestige * RM	$b = 1.49$, 89% CrI:	$b = 0.400$, 89% CrI:	OR = 1.492, 89% CrI:	$b = 0.321$, 95% CI:	M20

		[1.20, 1.88]	[0.181, 0.632]	[1.198, 1.881]	[0.097, 0.545]	
	Formidability * RM (homicide rates high)	$b = 1.49$, 89% CrI: [1.12, 1.97]	$b = 0.400$, 89% CrI: [0.111, 0.683]	OR = 1.492, 89% CrI: [1.118, 1.980]	$b = 0.410$, 95% CI: [0.141, 0.680]	M57
	Prestige * RM (homicide rates high)	$b = 1.35$ 89% CrI: [1.07, 1.68]	$b = 0.300$, 89% CrI: [0.066, 0.518]	OR = 1.350, 89% CrI: [1.068, 1.678]	$b = 0.279$, 95% CI: [0.084, 0.473]	M58
Women's perception of male voices	Long term attractiveness	$b = 1.86$, 89% CrI: [1.39, 2.39]	$b = 0.616$, 89% CrI: [0.190, 0.946]	OR = 1.852, 89% CrI: [1.209, 2.576]	$b = 0.689$, 95% CI: [0.570, 0.808]	M1
	Short term attractiveness	$b = 1.57$, 89% CrI: [1.17, 2.03]	$b = 0.450$, 89% CrI: [0.021, 0.773]	OR = 1.568, 89% CrI: [1.021, 2.167]	$b = 0.523$, 95% CI: [0.404, 0.642]	M1
	Long term vs short term	OR = 1.18, 89% CrI: [1.10, 1.26]	OR = 1.189, 89% CrI: [1.089, 1.272]	-	OR = 1.181, 95% CI: [1.057, 1.319]	M1
	Short term * homicide rates	$b = 1.23$, 89% CrI: [1.04, 1.21]	$b = 0.120$, 89% CrI: [0.043, 0.191]	OR = 1.127, 89% CrI: [1.043, 1.210]	$b = 0.116$, 95% CI: [0.024, 0.209]	M62
	Long term attractiveness * homicide rates	$b = 0.95$, 89% CrI: [0.81, 1.12]	$b = -0.049$, 89% CrI: [-0.210, 0.107]	OR = 0.952, 89% CrI: [0.810, 1.113]	$b = -0.058$, 95% CI: [-0.256, 0.140]	M62
	Short term attractiveness *	$b = 1.07$, 89% CrI: [-	$b = 0.066$, 89% CrI: [-	OR = 1.068, 89% CrI: [-	$b = 0.058$, 95% CI: [-	M62

	homicide rates	[0.90, 1.26]	0.097, 0.225]	0.908, 1.252]	0.140, 0.257]	
	Slope differences: short term vs long term	$b = 0.89$, 89% CrI: [0.83, 0.96]	$b = -0.115$, 89% CrI: [-0.191, -0.043]	OR = 0.891, 89% CrI: [0.826, 0.958]	$b = -0.116$, 95% CI: [-0.209, -0.024]	M62
Men's perception of female voices	Short term attractiveness	$b = 0.78$, 89% CrI: [0.59, 1.03]	$b = -0.252$, 89% CrI: [-0.589, 0.164]	OR = 0.777, 89% CrI: [0.556, 1.178]	$b = -0.263$, 95% CI: [-0.382, -0.145]	M1
	Long term attractiveness	$b = 1.51$, 89% CrI: [1.16, 2.01]	$b = 0.409$, 89% CrI: [0.051, 0.808]	OR = 1.505, 89% CrI: [1.052, 2.244]	$b = 0.404$, 95% CI: [0.285, 0.523]	M1
	Short term vs long term	OR = 1.94, 89% CrI: [1.83, 2.06]	OR = 1.937, 89% CrI: [1.794, 2.087]	-	OR = 1.949, 95% CI: [1.751, 2.169]	M1
	Short term * RM	No effect in text	$b = 0.090$, 89% CrI: [-0.120, 0.298]	OR = 1.094, 89% CrI: [0.887, 1.347]	$b = 0.105$, 95% CI: [-0.037, 0.246]	M18
	Long term * RM	No effect in text	$b = 0.240$, 89% CrI: [-0.036, 0.526]	OR = 1.271, 89% CrI: [0.964, 1.691]	$b = 0.178$, 95% CI: [-0.036, 0.392]	M17
Women's perception of female voices	Flirtatiousness	$b = 0.73$, 89% CrI: [0.57, 0.98]	$b = -0.305$, 89% CrI: [-0.658, 0.098]	OR = 0.737, 89% CrI: [0.518, 1.102]	$b = -0.317$, 95% CI: [-0.435, -0.199]	M1
	Attractiveness to men	$b = 0.90$, 89% CrI: [0.69, 1.20]	$b = -0.110$, 89% CrI: [-0.439, 0.327]	OR = 0.896, 89% CrI: [0.645, 1.386]	$b = -0.121$, 95% CI: [-0.239, -0.003]	M1

	Flirtatiousness * RM	$b = 1.55$, 89% CrI: [1.16, 2.03]	$b = 0.44$, 89% CrI: [0.149, 0.710]	OR = 1.553, 89% CrI: [1.161, 2.033]	$b = 0.470$, 95% CI: [0.269, 0.671]	M23
Note: RM stands for relational mobility; CrI stands for credible interval.						